DEVOPS – WEEKLY E12 NOTES

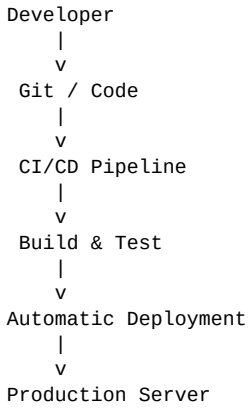
5–6 Mark Answers
Difference questions contain 8 easy points. Important diagrams are included.

1. Automated and Manual Deployment – Difference

Theory: Automated deployment uses tools and scripts to deploy an application automatically. Manual deployment is performed by a person by carrying out deployment steps such as copying files, configuring servers and restarting services.

Automated Deployment	Manual Deployment
Deployment is performed automatically using tools/scripts.	Deployment is performed manually by a person.
Requires less human effort.	Requires more human effort.
Faster deployment.	Usually takes more time.
Less chance of human error.	Higher chance of human error.
Easily repeatable.	Repeating steps can be difficult.
Suitable for frequent deployments.	Suitable for occasional/small deployments.
Can be integrated with CI/CD pipelines.	Usually requires manual intervention.
Provides consistent deployment.	Results may differ because of manual steps.

Diagram:

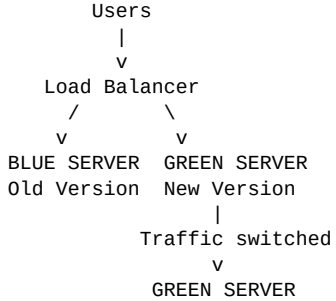


Example: Jenkins, GitHub Actions and GitLab CI/CD can be used for automated deployment.

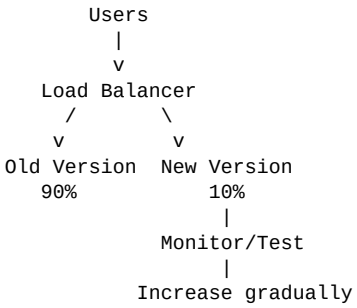
2. Blue-Green and Canary Deployment

Theory: Blue-Green and Canary are deployment strategies used to release new application versions with reduced deployment risk.

Blue-Green Deployment: Two environments are maintained. Blue contains the current version and Green contains the new version. After testing, traffic is switched to Green.



Canary Deployment: The new version is first given to a small percentage of users. If it works correctly, traffic is gradually increased.



Blue-Green	Canary
Uses two complete environments.	Uses controlled traffic to the new version.
One environment has the old version.	A small group gets the new version.
Traffic is switched between environments.	Traffic is gradually increased.
Usually needs duplicate infrastructure.	Does not necessarily need a complete duplicate environment.
Rollback is done by switching traffic back.	Rollback is done by reducing/removing new-version traffic.
New version can be tested before switching users.	New version is tested with real users gradually.
Deployment switch can be quick.	Deployment is gradual.
Useful when a parallel environment is available.	Useful when gradual real-user testing is desired.

3. How Kubernetes Works in Automating Software Deployment

Theory: Kubernetes is an open-source container orchestration platform. It automates deployment, scaling and management of containerized applications.

Working:

• Developer creates the application and packages it into a container.

• Container image is stored in a container registry.

• Kubernetes receives a deployment configuration.

• The Control Plane manages the Kubernetes cluster.

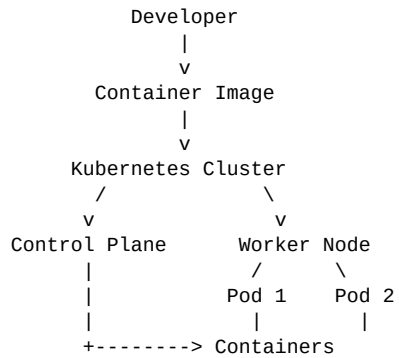
• Worker Nodes run application containers inside Pods.

• Kubernetes maintains the desired number of Pods.

• If a Pod fails, Kubernetes can create a replacement.

• Kubernetes can scale applications according to requirements.

Diagram:



Main components: Control Plane, API Server, Scheduler, Pod, Worker Node and Service.

Result: Kubernetes automates application deployment, scaling, load distribution and recovery of failed containers.

4. CI/CD Build Process

Theory: CI/CD stands for Continuous Integration and Continuous Delivery/Deployment. It automates building, testing and delivering software when developers make changes.

Process:

• Developer writes or modifies code.

• Code is pushed to a Git repository.

• CI/CD pipeline is triggered.

• Source code and dependencies are prepared.

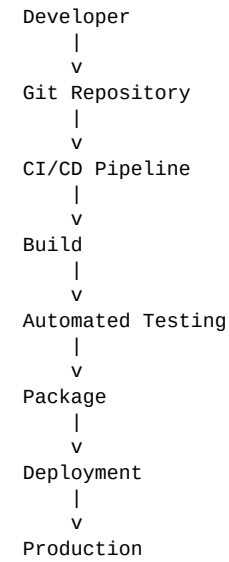
• Application is compiled/built.

• Automated tests are executed.

• If tests pass, the application is packaged.

• The application is deployed to the required environment.

Diagram:



Examples: Jenkins, GitHub Actions, GitLab CI/CD and Azure DevOps.

Advantages: Faster delivery, automated testing, fewer manual errors, early bug detection and reliable releases.

5. Role of DevOps Engineer in Managing Infrastructure as Code (IaC)

Theory: Infrastructure as Code means managing servers, networks, databases and other infrastructure using configuration files or code instead of manual configuration.

Roles:

• Creates infrastructure configuration files.

• Automates server and resource creation.

• Maintains infrastructure configurations.

• Uses version control such as Git.

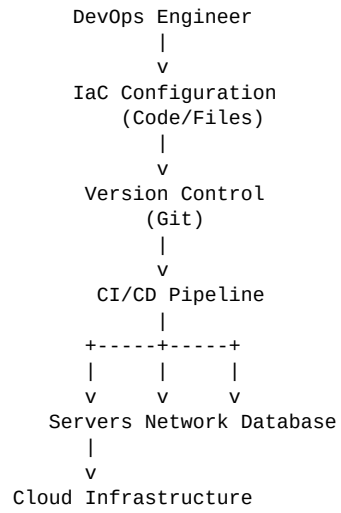
• Integrates IaC with CI/CD pipelines.

• Monitors and updates infrastructure.

• Maintains consistency between environments.

• Helps scale infrastructure automatically.

Diagram:



Common tools: Terraform, Ansible and AWS CloudFormation.

Example: Instead of manually creating a server, a DevOps engineer can define the server configuration in a Terraform file and create it automatically.

6. Automated and Manual Testing – Difference

Theory: Manual testing is performed by a tester manually executing test cases. Automated testing uses software tools and scripts to execute test cases automatically.

Manual Testing	Automated Testing
Tests are performed manually by testers.	Tests are performed using scripts/tools.
Requires more human effort.	Requires less repeated human effort.
Takes more time for repeated tests.	Repeated tests are faster.
Human errors are possible.	Reduces human errors in repetitive execution.
Programming is not always required.	Usually requires test scripts/programming.
Suitable for exploratory and usability testing.	Suitable for repetitive and regression testing.
Difficult to run the same test many times quickly.	Same tests can be executed repeatedly.
Results may need to be recorded manually.	Results can be generated automatically.

Diagram:

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Manual:
Tester -> Test Case -> Manual Execution -> Result

Automated:
Test Script -> Testing Tool -> Automatic Execution -> Result
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Example: Manual testing checks a login form by a tester. Automated testing can use a Selenium script to enter username/password and verify the result.

Quick Revision

1. Automated vs Manual Deployment – 8 differences
2. Blue-Green vs Canary – definitions + 8 differences + diagrams
3. Kubernetes – working + components + diagram
4. CI/CD – process + flow diagram
5. DevOps Engineer & IaC – roles + workflow diagram

6. Manual vs Automated Testing – 8 differences + diagram